Songhang Li

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Education

Columbia University — New York, NY
M.S. in Electrical Engineering — *May 2028*
B.S. in Electrical Engineering — *May 2027*

Franklin & Marshall College — Lancaster, PA
B.A. in Physics and Mathematics | *Magna Cum Laude* | ΦBK — *May 2025*

Academic & Research Experiences

Digital Very Large-Scale Integration — Sep. 2026 – Present
Digital IC Designer *Advisor: Prof. Ken Shepard* — *Columbia University, NY*

- Designing an 8-bit microcontroller in TSMC 65-nm CMOS, integrating SRAM memory, accumulator, adder, shifter, decoder, MUX, and bus driver interface.
- Developing schematics and layout using Composer and Virtuoso, integrating the datapath, PLA, and power grid.
- Verifying layout through DRC and LVS checks in Mentor Calibre, with parasitic extraction using xRC.
- Optimizing design by considering ESD protection, boundary scan, inductance, synchronization.
- Evaluating power, timing, and area tradeoffs, including latch and clocking design, noise, testability, and reliability.

Analog Electronic Circuits - Analog Computer & Amplifier Design — Sep. 2026 – Present
Analog Designer *Advisors: Subhajit Ray, Prof. Peter Kinget* — *Columbia University, NY*

- Developing an AD8542 op-amp-based analog computer in Cadence Virtuoso to model a mass–spring system using integrator and inverting amplifier circuits.
- Solving coupled second-order differential equations via integrator and inverting op amp configurations.
- Mapping coupled second-order differential equations to circuit (integrators and inverting op-amps) implementations and evaluating transient responses under varied initial capacitor voltages.
- Characterizing op-amp circuit topologies through DC, AC, and transient simulations to evaluate operating points, gain, and frequency response.
- Designing and analyzing MOS amplifiers and differential input stages, evaluating transistor sizing, small-signal model, current mirror, feedback stability, compensation, slew rate, and offset.

Laser Induced Transfer Forwarding vs E-beam evaporation on PVDF — May 2026 – Present
Undergraduate Researcher *Advisor: Prof. Ioannis Kymissis, Peter Ballentine* — *Columbia University, NY*

- Investigating LIFT as a low-thermal-budget to E-beam evaporation for PVDF electrode deposition, evaluating its ability to preserve piezoelectric performance(β-phase) through comparative characterization.
- Developing a data acquisition system integrating an oscilloscope and ADMET EP1 electromechanical controller to quantify the relationship between applied force rate and PVDF charge output.
- Designing electrode and interconnect layouts in KLayout and preparing vinyl stencil.
- Fabricating screen-printed silver-paste electrodes, enabling low-temperature processing to preserve PVDF piezoelectric properties.

IEEE SSCS 2026 Chipathon Integrated Circuit Design Contest — May 2026 – Present
Analog IC Designer *Advisor: Prof. Peter Kinget, Luighi Viton-Zorrilla* — *Columbia University, NY*

- Designing a programmable five-transistor operational transconductance amplifier for the open-source Four-Topology MOSbius test chip in the GF180MCU 180 nm process.
- Applying gm/Id-based transistor sizing and Xschem/ngspice simulations to evaluate gain, bandwidth, stability, slew rate, power consumption, and output range across programmable sizing modes.
- Completing the OTA layout in KLayout with DRC and LVS clean verification.
- Preparing tapeout by simulating inverting op amp implementation for top-level verification.
- Preparing a manuscript for submission in *MIT Undergraduate Research Technology Conference* and *IEEE Xplore*.

Photon Counting Experiment of Persistent Luminescence in Eu:LAP — May 2024 – Jan. 2025
Research Assistant *Advisor: Prof. Ken Krebs* — *F&M, Lancaster, PA*

- Measured decay dynamics of excited population of impurity ions after UV laser excitation using photon counting.

- Developed a circuit solution using Arduino to precisely control excitation laser mechanical shutter, enabling photon counting of early state decay.
- Compared the emission spectra measurement with calculated energy levels of Eu^{3+} in $LaAlO_3$ using density functional theory.
- Conducted a literature review and originated a table summarizing experimental technique and materials.
- Prepared manuscript, see publication (Materials Letters, 414.)

Society of Automotive Engineers Aero Design Competition Preparation Sept. 2023 – May 2025
Team Lead, Pilot Advisor: Prof. Ken Krebs *F&M, Lancaster, PA*

- Designed a power and lift efficient, short-take-off-length 3.8 meters fixed wing plane for water delivery.
- Calculated the lift of the wing based on the structure of air foil in order to improve the load capacity.
- Fabricated the wings utilizing balsa wood and carbon fiber, employing laser cutter and heat-shrink covering film application techniques.
- Led the team members to write the design report and presented the result in a research fair.
- Wrote a funding proposal and managed to receive a year-long student research grant from the grant committee.

Universal Village Program-Smart Materials and Intelligent Electronics Jun. 2022 – Aug. 2022
Research Assistant Advisor: Dr. Lin Zhang, Massachusetts Institute of Technology Media Lab *Boston, MA*

- Designed a flexible electrode pattern exposure mask using AutoCAD, in preparation for printing of mask.
- Prepared piezoelectric 1-3 composites for piezoelectric sensors.
- Organized the piezoelectric hydrogel sensors comparison table and contributed to manuscript writing, see publication (Micromachines 2023, 14, 167).

Publications

- J.K. Krebs, **Songhang Li**, Dragos Georgescu, and T. D. Pham. (2026). *Eu3+ Luminescence as a Probe of Local Crystallinity in Combustion-Synthesized LaAlO3. Materials Letters*, 414. doi:10.1016/j.matlet.2026.140581
- Eli Johnson, Max Drucker, Manuel Garcia, **Songhang Li** (2026). *A Programmable Four-Topology Amplifier Chip for Analog Education. MIT URTC | IEEE Xplore* in prep.
- Yuxuan Du, Wenya Du, Dabin Lin, Minghao Ai, **Songhang Li**, Lin Zhang. (2023). *Recent Progress on Hydrogel-Based Piezoelectric Devices for Biomedical Applications. Micromachines*,14(1). doi:10.3390/mi14010167

Awards & Honors

Frank Durrell Enck Memorial Prize in Physics | *F&M* Spring 2025
- Established in 1975; traditionally awarded to a graduating Physics major whose overall departmental performance and involvement is most deserving of recognition.

Kershner Scholar in Mathematics | *F&M* Spring 2025
- Awarded to students based on their proficiency in Mathematics.

Phi Beta Kappa (ΦBK) | *U.S.A.* Since 2025
- Invitation-only academic honor society recognizing excellence in the liberal arts and sciences.

The Nolt Music Award | *F&M* Spring 2025
- Awarded to F&M student musicians to undertake musically enriching projects such as a senior recital.

The Alma Mater Singer | *F&M (238th Commencement)* Spring 2025
- Selected as the Alma Mater Singer for Franklin & Marshall College's 238th Commencement.

Summer Scholar Award | *F&M* Summer 2024
- Awarded to a student researcher for summer research conducted with faculty.

Pi Mu Epsilon (Math Honor Society) | *U.S.A.* Since Spring 2024
- Nationally recognized honorary society dedicated to furthering knowledge of mathematics.

Sigma Pi Sigma (Physics & Astronomy Honor Society) | *U.S.A.* Since Spring 2023
- Honors outstanding scholarship in physics and astronomy.

Student Independent Research Grants | *F&M* Fall 2023
- Awarded for advanced student research/projects whose needs exceed departmental budgets.

Kershner Scholar in Physics | *F&M* Spring 2023
- Awarded to students based on their proficiency in Physics.

Dean's and Honors Lists | *F&M* Since Fall 2021
- Earned Honors List recognition for achieving a 3.7 or better grade point average.

Teaching Experience

Department of Electrical Engineering Feb. 2026 – May 2026
Course Assistant *Columbia University, New York, NY*

- Prepared the LED matrix via passive scanning using Arduino for ENGI E1102: *The Art of Engineering.*

Department of Physics and Astronomy Jan. 2024 – May 2025
Teaching and Research Assistant *Franklin & Marshall College, Lancaster, PA*

- Compared two electronics textbooks and exercises to identify key instructional concepts for Intro to Electronics.
- Tested different teaching kits selected by the professor and evaluated suitability for effective teaching and learning.

Presentations & Conferences

MIT Undergraduate Research Technology Conference, Paper Presentation | *MIT, MA* Oct. 10, 2026

- A Programmable Four-Topology Amplifier Chip for Analog Education.

American Physical Society Meeting, Mid-Atlantic Section | *Temple University, PA* Nov. 15, 2024

- Rate equation model for fluorescence and phosphorescence in Eu^{3+} doped LaAlO3.

Student Research Fair | *Franklin and Marshall College, PA* Oct. 18, 2024

- Photon Counting Experiment of Persistent Luminescence in Eu:LAP

Student Research Fair | *Franklin and Marshall College, PA* Apr. 20, 2024

- Design, construction, testing, and improving: An introduction to Engineering Design Competitions.

Skills & Instrumentation

EDA Tools & Design Software: Cadence(Composer, Virtuoso, Spectre), Mentor(Calibre, xRC),LT SPICE, NG SPICE, MATLAB, Xschem, KLayout, Arduino IDE, Onshape.
Processes: Global Foundries GF 180nm, TSMC 65nm.
Hardware & Instrumentation: ADMET Testing System, Oscilloscopes, Waveform Generators, HP LCR, M-2000 J.A. Woollam Spectrum ellipsometry, Monochrometer SR430, Stanford Research System Multichannel Photon Counter.
Programming language: Python, Java, MATLAB, MIPS Assembly, Mathematica, R.
Software: IntelliJ, AutoCAD, MATLAB, CompleteEASE(optical modeling and ellipsometry data fitting)
Design & Analysis Skills: Electromechanical characterization, Analog and digital circuit simulation (AC/DC/transient analysis), Circuit design and debugging, Data analysis with Python/MATLAB/SciDavis, Characterization of electronic components and semiconductor materials.
Additional Technical Skills: Laser cutting/fabrication (Glowforge), Vinyl cutter, 3D printing, PCB prototype including exposure and etching technics, technical documentation (LaTeX, Endnote), design visualization and project planning (X-mind), image processing (Adobe Lightroom, Photoshop), Screen printing.

References

Ioannis Kymissis Kenneth Brayer Professor of Electrical Engineering, Columbia University
Charles Zukowski Professor of Electrical Engineering, Former EE Department Chair, Columbia University
Ken Krebs Professor of Physics, Franklin & Marshall College
Lin Zhang Asistant Professor of Biomedical Engineering,University at Buffalo The State University of New York
Calvin Stubbins Professor of Physics, Franklin & Marshall College
Micheal McCooey Assosiate Professor of Mathematics, Franklin & Marshall College
Savannah Eisner Assistant Professor of Electrical Engineering, Columbia University

Coursework

Electrical Engineering: Digital VLSI, Analog Electronic Circuit, Microfabrication, Solid State Devices & Material, Signal & System, Communication System, Modern Display Science & Technology.
Physics: Advanced Methods in Physics, Electromagnetism, Quantum Mechanics, Analytic Mechanics, Mathematic Methods in Physics, Statistical & Thermal Physics.
Math: Advanced Linear Algebra, Abstract Algebra, Multi-variable Calculus, Probability & Statistics, Differential Equation, Origami Math, Real Analysis.